

DNA splice site sequences clustering method for conservativeness analysis

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Abstract

DNA sequences that are near to splice sites have remarkable conservativeness, and many researchers have contributed to the prediction of splice site. In order to mine the underlying biological knowledge, we analyze the conservativeness of DNA splice site adjacent sequences by clustering. Firstly, we propose a kind of DNA splice site sequences clustering method which is based on DBSCAN, and use four kinds of dissimilarity calculating methods. Then, we analyze the conservative feature of the clustering results and the experimental data set.

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1. Introduction

Human Genome Project (HGP) aimed at elucidating the sequence of human being's three billion base pairs, discovering all the genes and their positions on chromosomes, and decrypting all the human being's genetic information. Along with the successful completion of HGP, bioinformatics comes into the post-genome era. Now, mining knowledge from the mass data has become much more important than obtaining biological data. Many studies are concentrated on the classification of sequences, discriminating coding section from non-coding section, structure prediction, function prediction, and so on. The main methods that researchers often use include neural network method [1], Bayes algorithm [2], HMM [3] (hidden Markov models), SVM [4] (support vector machine), and clustering methods [5].

The motive for clustering analysis is to divide the objects into several clusters; a cluster is a collection of objects that are similar to each other, and objects in different clusters are dissimilar from each other. Clustering is used in many applications such as text mining [6], web data analysis [7], and image processing [8]. Because similar DNA sequences usually have similar functions, researchers often predict function by sequence alignment. Therefore, in this study, we present a DBSCAN-based splice site sequences clustering method to find the feature of splice site sequences, which will be helpful to predict their functions. DBSCAN is chosen as the clustering method because it is better in the following aspects than K-means, which is one popular method in the field of bioinformation processing: (i) K-means must determine a cluster number before clustering, while DBSCAN does not need it, DBSCAN finds dense clusters automatically for a given density threshold; (ii) K-means is sensitive with noises, but DBSCAN handles noises well; (iii) K-means is only applicable to find circular or spherical clusters, but DBSCAN can find clusters with arbitrary shapes; (iv)

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K-means has the numeric-only limitation, but DBSCAN can work well with categorical data.

In this paper, we will exhibit some dissimilarity calculating methods between DNA sequences, and propose one new kind of dissimilarity calculating method. Then, after clustering, we analyze the conservative characterization of the clustering result and the experimental data set.

2. Cluster-based bioinformation processing

Clustering plays an important role in the field of bioinformation processing, and it is widely used in gene database knowledge discovery. It is used mainly in the following fields:

- (1) Clustering can effectively partition different DNA sequence sets into different clusters by sequence alignment. Based on the clusters, further research can be developed. For example, Michael Eisen developed one microarray data analysis software called CLUSTER [9]. CLUSTER finds the most relevant gene pair by comparing genes with each other. Then, the average of the gene pair substitutes the gene pair, and the relevant matrix is calculated again. Repeat the process until the result is satisfactory.
- (2) Clustering is applied in the field of function prediction. Several clustering methods are proposed to predict the secondary structures of RNA and protein [10]. Because the functions and the structures are closely related, these clustering methods are helpful to predict their functions. Clustering also does well in the identification and prediction of exons, introns and splice sites [11].
- (3) Clustering also plays an important role in the research of molecule evolution theory [12]. Because evolution tree is the main method for analyzing molecule evolution, it provides an ideal opportunity for hierarchical clustering methods. Evolution trees are constructed by the dissimilarity between DNA sequences or protein structures, so it is obvious that clustering is one useful tool to construct evolution trees.

In brief, comparison of dissimilarity between objects is the main method for mining knowledge in gene database, and so clustering is an indispensable method in the field of biological knowledge discovery.

3. Splice site adjacent sequences clustering

3.1. DBSCAN method description

DBSCAN (density-based spatial clustering of applications with noise) is a density-based algorithm. DBSCAN is based on the fact that clusters are of higher density than its surroundings. The key idea of DBSCAN is that for each object of a cluster, the neighborhood with a given radius ε

must contain at least a minimum number of *MinPts* objects, i.e. the cardinality of the neighborhood has to exceed a given threshold. In what follows, we will present the basic definitions of DBSCAN [13].

Definition 1 (*neighborhood*). The neighborhood of object p is the set of objects in the circle, whose center is p and radius is ε , i.e. $N_\varepsilon(p) = \{q \in D | \text{dist}(p, q) \leq \varepsilon\}$, where D is the database of objects.

Definition 2 (*directly density reachable*). An object p is directly density reachable from an object q wrt ε and *MinPts* if p is within the neighborhood of q , i.e. $p \in N_\varepsilon(q)$; and q is core object, i.e. $|N_\varepsilon(q)| \geq \text{MinPts}$ (core object condition).

Definition 3 (*density reachable*). An object p is density reachable from an object q wrt ε and *MinPts* if there is a chain of objects p_i ($i = 1, \dots, n$), and p_i is directly density reachable from p_{i+1} ; p is p_1 , and q is p_n .

Definition 4 (*density connected*). An object p is density connected to another object q wrt ε and *MinPts* if there is another object p_t such that both p and q are density reachable from p_t wrt ε and *MinPts*.

Definition 5 (*noise*). Let $C_1, \dots, C_k$ be the clusters wrt ε and *MinPts*. If p is an object of D , and it does not belong to any cluster C_i ($i = 1, \dots, k$), then p is a noise, i.e. $\text{noiset} = \{p \in D | \forall i : p \notin C_i\}$, ($i = 1, \dots, k$).

A cluster is one set of objects, which are density connected with each other. The algorithm is given as follows.

Algorithm DBSCAN

Input: objects $D = \{p_i\}$ ($i = 1, \dots, N$), ε , *MinPts*.

Output: clusters and noise set.

① Check the object p that has not yet been processed (clustered or marked as noise). If p is a core object, a new cluster C is created, the neighbors of p which are not yet contained in any cluster are added into cluster C .

② Check the object q in C which has not been checked, if q is a core object, the neighbors of q which are not yet contained in any cluster are added into cluster C .

③ Repeat ② until all the objects in C are checked.

④ Repeat ①, ② and ③ until all the objects are classified as some cluster or noise.

3.2. Dissimilarity definitions

The definition of dissimilarity between objects is crucial for clustering, and now we will discuss some common dissimilarity definitions between DNA sequences and then introduce one dissimilarity definition that is proposed in this study.

3.2.1. Direct alignment

Given two DNA sequences $X = (x_1, x_2, \dots, x_n)$ and $Y = (y_1, y_2, \dots, y_n)$, at first, we compare each base pair at the corresponding position directly. If the base pair is the

same the dissimilarity between them is 0, otherwise the dissimilarity between them is 1. Then, summate the dissimilarity of each base pair, and the sum is the dissimilarity between X and Y . The computation formula is as follows:

$$Dist(X, Y) = \sum_{i=1}^n \sigma(x_i, y_i) \quad (1)$$

$$\text{where } \sigma(x_i, y_i) = \begin{cases} 0, & x_i = y_i \\ 1, & x_i \neq y_i \end{cases}$$

Direct alignment is simple, but the sequences must be with the same length.

3.2.2. Dynamic programming alignment

In the evolution process of biology, gene mutation may take place at some local areas of DNA sequence. The types of mutations include substitution, insertion, deletion, and rearrangement.

Dynamic programming alignment method does well with gene mutations, which can find out the optimal alignment result between sequences. The standard computation consists of aligning one sequence with the other according to a cost function computed optimally by dynamic programming. An alignment of sequences $X = (x_1, x_2, \dots, x_n)$ and $Y = (y_1, y_2, \dots, y_m)$ defines a sequence of edit operations that transforms Y into X by matching, substituting, inserting, or deleting bases in X and Y on a base-by-base basis. Each individual edit is assigned a real-valued cost by the cost function, and the costs in a complete edit sequence are added. An optimal alignment of X and Y is one, for which the edit sequence has the minimum total cost. The computation is organized by systematically filling in a dynamic programming matrix with time and space complexity $O(mn)$.

Smith–Waterman algorithm is one well-known dynamic programming algorithm. The process of Smith–Waterman algorithm is as follows.

Algorithm Smith–Waterman

Input: $X = (x_1, x_2, \dots, x_n)$, $Y = (y_1, y_2, \dots, y_m)$.

Output: optimal alignment between X and Y .

Calculate the dissimilarities between the sequences X and Y , and then save the dissimilarities in the edit matrix, then find the optimal alignment between X and Y by back-tracking method according to the edit matrix.

Prior conditions:

$$V(0, 0) = 0$$

$$V(i, 0) = V(i - 1, 0) + \sigma(X[i], -)$$

$$V(0, j) = V(0, j - 1) + \sigma(-, Y[j])$$

Recursion conditions:

$$V(i, j) = \min \begin{cases} V(i - 1, j - 1) + \sigma(X[i], Y[j]) \\ V(i - 1, j) + \sigma(X[i], -) \\ V(i, j - 1) + \sigma(-, Y[j]) \end{cases}$$

Here, $\forall x, y (x \neq y), \sigma(x, x) = 0,$

$$\sigma(x, y) = \sigma(x, -) = \sigma(-, y) = 1.$$

3.2.3. Statistical probability-based method

The percentage of various bases contained in sequences which have different functions will be different, and so researchers often classify DNA sequences by the statistical characterization. Because the value of statistical probability is interval data, Euclidean distance can be used to present the dissimilarity between sequences.

Firstly, calculate the percentage of various bases contained in each sequence. For example, the statistical results of sequence $X = (x_1, x_2, \dots, x_n)$ and $Y = (y_1, y_2, \dots, y_n)$ are $F_{X_h}, F_{Y_h} (h \in \{A, T, G, C\})$.

Secondly, standardize the data by the following steps.

Suppose that the number of the sequences is N , and the percentage of base h contained in the I th sequence is $F_{Ih} (h \in \{A, T, G, C\}, I = 1, 2, \dots, N)$.

① Calculate the mean absolute deviation S_h

$$S_h = (|F_{1h} - m_h| + |F_{2h} - m_h| + \dots + |F_{Nh} - m_h|) / N$$

Here, $m_h = (F_{1h} + F_{2h} + \dots + F_{Nh}) / N, h \in \{A, T, G, C\}$.

② Calculate the standardized values F'_{Ih}

$$F'_{Ih} = (F_{Ih} - m_h) / S_h, (h \in \{A, T, G, C\}, I = 1, 2, \dots, N).$$

③ Calculate dissimilarities between sequences.

Suppose that the standardized values of X and Y are $F'_{X_h}, F'_{Y_h} (h \in \{A, T, G, C\})$. Then, the dissimilarity between X and Y is

$$Dist(X, Y) = \sqrt{\sum_{h \in \{A, T, G, C\}} (F'_{X_h} - F'_{Y_h})^2}$$

Besides the method mentioned above, researchers also use the statistical probability of base combinations to calculate dissimilarity between DNA sequences, such as the statistical probability of purine (A and T), or the statistical probability of various codons.

3.2.4. Direct alignment in combination with statistical information

The statistical characterization of splice site adjacent sequences is remarkable. The occurrence frequencies of various bases are different at different positions on the sequences. This is very important information about the sequences. But when dissimilarity between sequences is calculated by direct alignment, all the mismatching cases are regarded as the same, and the statistical characterization is ignored unfortunately. So we propose a dissimilarity definition method of direct alignment in combination with statistical information.

Suppose that the number of the sequences is N , then the dissimilarity between sequences $X = (x_1, x_2, \dots, x_n)$ and $Y = (y_1, y_2, \dots, y_n)$ can be calculated by the following functions:

$$\begin{cases} \mu(x, y) = \begin{cases} 0, & x = y \\ 1, & x \neq y \end{cases} \\ \sigma(x_i, y_i) = (|f[x_i] - f[y_i]| + 1) * \mu(x_i, y_i) \\ Dist(X, Y) = \sum_{i=1}^n \sigma(x_i, y_i) \end{cases} \quad (2)$$

where

$$f[x_i] = (\text{the occurrence times of } x_i \text{ at the } i\text{th position})/N,$$

$$f[y_i] = (\text{the occurrence times of } y_i \text{ at the } i\text{th position})/N.$$

It is easy to verify that if x_i is the same as y_i , the result will be the same as direct alignment; but if x_i does not match with y_i , there will be some difference, and the result will be $\sigma(x_i, y_i) = |f[x_i] - f[y_i]| + 1$. How to explain it from the aspect of biology? We found that at many positions of splice site adjacent sequences, the occurrence probability of some bases is much greater than that of the others. In this case, the base with much higher occurrence probability is correspondingly steady, and it is difficult for the base to mutate. In this situation, the value of $|f[x_i] - f[y_i]|$ as amending value presents how difficult for other bases to substitute the base.

4. Experimental results

We choose 2796 donor sequences of human being as the experimental data set (from HS3D, <http://www.sci.unisan-nio.it/docenti/rampone>). Fig. 1 shows the occurrence frequencies of various bases at each position with the length of 21 bits, and the 11th bit and 12th bit are both donor sites. It shows that the sequences from the 8th bit to the 17th bit have much more remarkable conservativeness, so we focus on these sequences (the length is 10 bits, and both the fourth bit and the fifth bit are donor sites).

Firstly, we cluster the sequences by DBSCAN, and then analyze the clustering results to mine the regulations in each cluster. Some analysis results are shown as follows.

4.1. Frequencies of various bases at each position

Dissimilarity definition formula: using (2);
parameters: $MinPts = 6$, $\varepsilon = 1.57$;
results: 4 clusters and 153 noise objects.

Fig. 2 shows that each cluster has much more apparent conservativeness than that in the experimental data set (Human being's 2796 donor sequences), and the conservativeness of different clusters are distinct. So we can predict

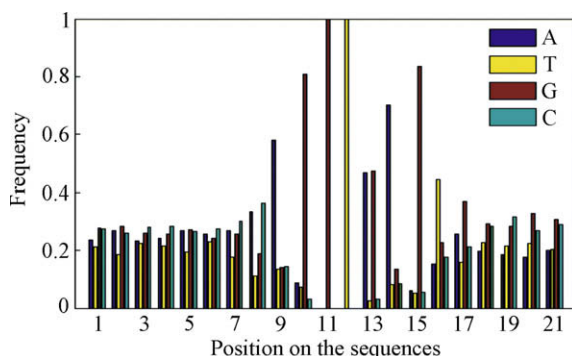


Fig. 1. Bases frequency distribution of donor site sequences.

that sequences in the same clusters may have more similar structures than those in different clusters.

4.2. Noise set analysis

Sometimes, we will pay more attention to the abnormal sequences than to the clustering results. The abnormal sequences maybe have some special structures and functions. In this study, we also analyze the noise set.

Dissimilarity definition formula: using (1);

parameters: $MinPts = 4$, $\varepsilon = 2.0$;

results: one cluster and 15 noise objects.

Table 1 shows the noise set.

Fig. 3 shows the occurrence frequencies of various bases at each position in noise set and the experimental data, and it shows us that the statistical feature of noise set is quite different with the feature of the experimental data.

4.3. Frequency of base combinations

Because adenine (A) and guanine (G) belong to the purine class base, which have similar structure and the other similar characters in certain aspects, (thymine (T) and cytosine (C) belong to the pyrimidine class base, and they also have some similar characters), the percentage of “T+C” (or “A+G”) is used to describe a given sequence sometimes. On the other hand, A and C have the same functional amino-group (both T and G have functional keto-group), and researchers also use “A+C” content to show the sequences’ characteristic. Some research results show that the sequences with different structures and functions usually have different structural and functional statistical features, and so these features are helpful to predict their structures and functions.

Fig. 4 gives the comparison result. It shows that different clusters share different percentages of “T+C” and “A+C”.

4.4. Di-bases bias

The sequences with different structures and functions usually have different di-bases (directly adjacent two bases) bias, so it shows us another direction to mine the sequences’ characteristics. Before showing the result, we will first define some notations.

Let f_x denote the frequency of the base X ($X \in \{A, T, G, C\}$), and f_{xy} mean the frequency of di-bases XY ($X, Y \in \{A, T, G, C\}$). Given a completely random sequence, if the adjacent bases are independent of each other, theoretically $f_{xy} = f_x \cdot f_y$. Therefore, we can use $P_{xy} = f_{xy} / (f_x \cdot f_y)$ to describe the bias of XY in the sequence, and the deviation of P_{xy} from 1 can be construed as a measure of XY bias [14].

Fig. 5 compares the bias of various di-bases among clusters and the experimental data set, and it shows that different clusters have different di-bases bias, which can help us to make a conjecture about their respective structures and functions.

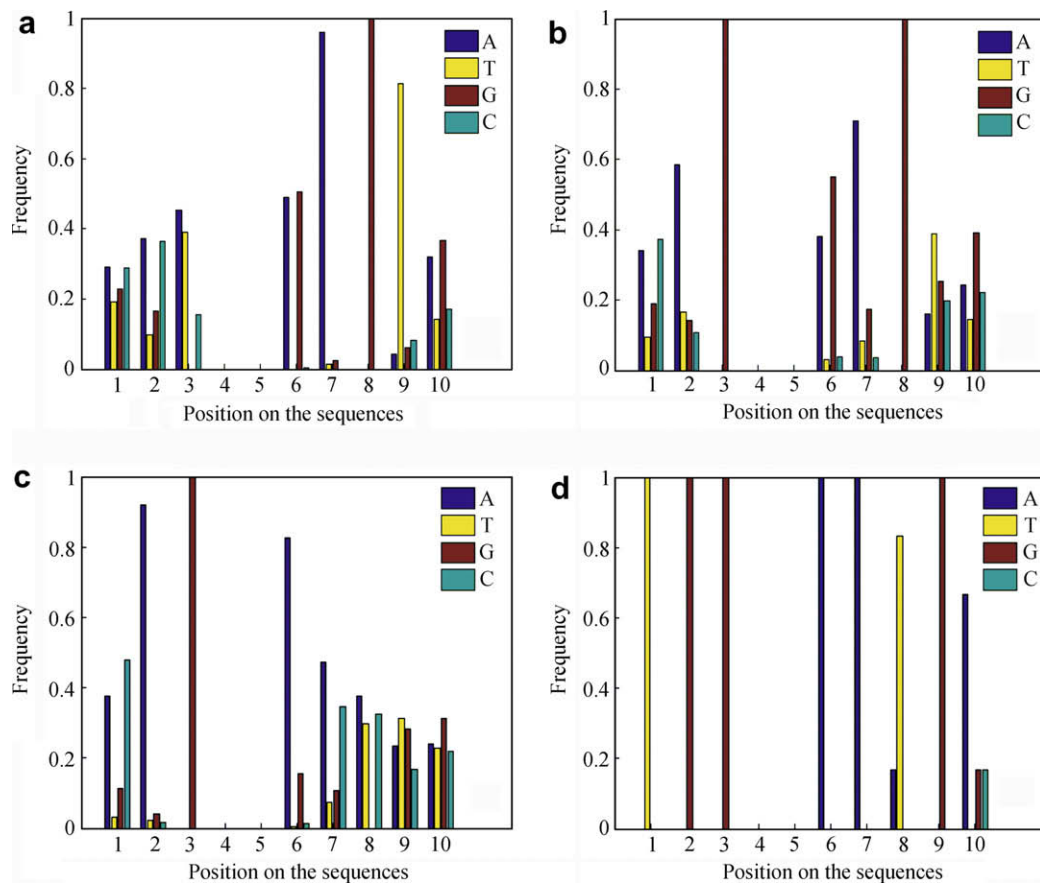


Fig. 2. Frequencies of various bases at each position in each cluster. (a) Cluster 1; (b) cluster 2; (c) cluster 3; (d) cluster 4. Because the donor sites of the sequences are all “GT”, the frequencies of the donor site are not shown.

Table 1

The noises wrt $MinPts = 4, \epsilon = 2.0$.

Sequences ^a (ID)	Sequences (ID)	Sequences (ID)
TGCGTTGAGT(36)	AGTGTTACTT(1213)	TGAGTCCCTG(1756)
AGAGTTCCTC(196)	GGAGTATCCT(1273)	AGAGTTAAGT(1773)
AAAGTTTATC(208)	ATTGTACATA(1310)	GAAGTTCTGG(2137)
TAAGTCAACG(723)	GCTGTATCCT(1494)	TGAGTCACCT(2138)
TAAGTCAACG(724)	TCAGTATACA(1495)	ACAGTGCTGG(2296)

^a Here, ID shows the id of the sequence in the HS3D donor data set, which are marked one by one from 1 to 2796.

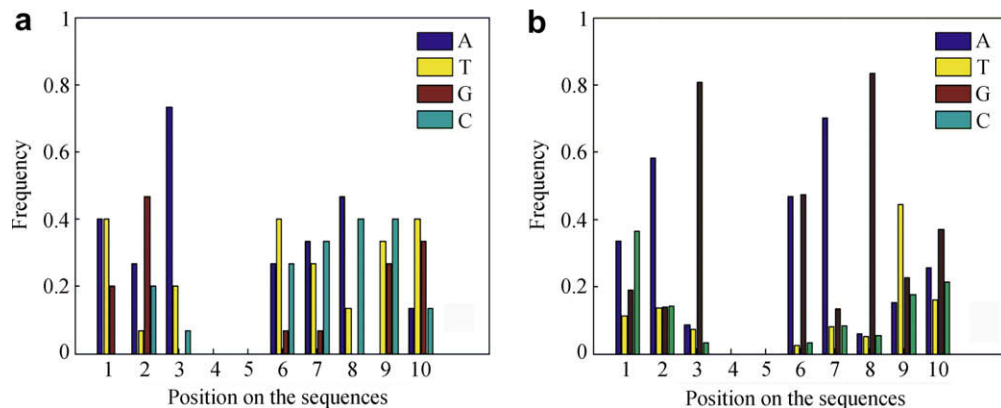


Fig. 3. Frequencies of various bases at each position. (a) The noise set; (b) the experimental data set.

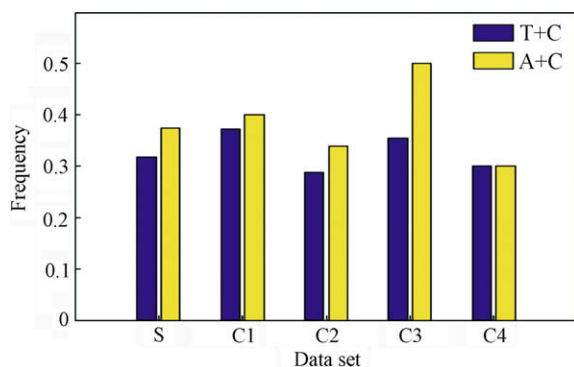


Fig. 4. Frequencies of $T+C$ and $A+C$ in the experimental data set and each cluster. S is the experimental data set, and C1, C2, C3, C4 mean cluster 1, cluster 2, cluster 3, cluster 4, respectively. Here, we use the clustering results in Section 4.1.

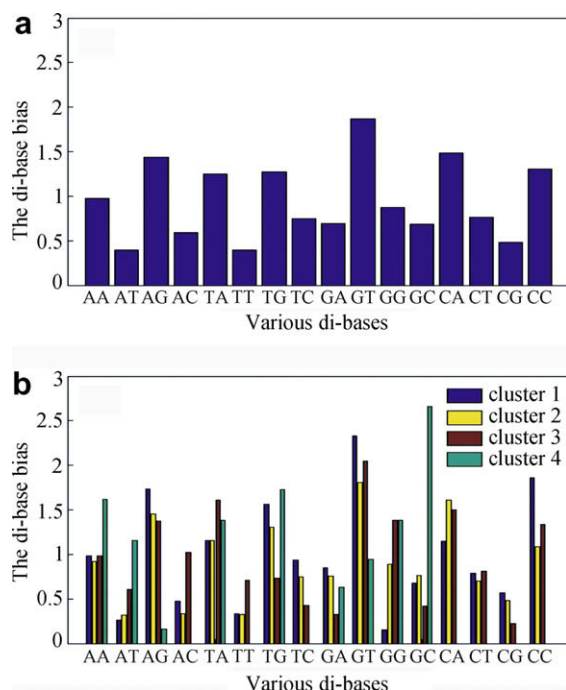


Fig. 5. The bias of various di-bases. (a) The bias (p_{xy}) of each di-base in the experimental data set; (b) the di-bases bias (p_{xy}) in each cluster. Here, we use the clustering results in Section 4.1.

5. Conclusions

The DNA splice site adjacent sequences have remarkable conservative feature, so there must be much genetic information. The research of DNA splice site sequences has become an important issue in the field of biological information processing, and many researchers have contributed to splice site prediction. In this study, we try to mine the underlying knowledge in DNA splice site sequences by clustering method.

Firstly, we cluster the DNA splice site sequences by DBSCAN. In order to improve the applicability of the algorithm, four kinds of dissimilarity definition methods are used. We can choose the proper definition method according to the aspects we care. Secondly, the bases' conservativeness of the

clusters is analyzed. Then, we compare the feature of noise sequence set with the feature of the experimental data set, and it shows us that they are quite different from each other. Lastly, the frequencies of " $T+C$ " ($A+C$) and the di-base bias are shown. These features can help us to predict the functions of the sequences in each cluster, and it will be also helpful to mine more biological knowledge from the clustering results.

Although the clustering results show us some important features, they do not provide enough biological knowledge directly. In the future, some further works will be done based on the clustering results to improve the prediction accuracy of splice sites.

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